Case Study: Replatforming an E-Commerce Database

Context

A retail engineering team is migrating their legacy PostgreSQL database to AWS Aurora to improve uptime and performance. The product manager has created an Epic named **[EPIC-102] Cloud Database Migration**. However, because database performance testing and infrastructure setup don't provide direct, standalone "features" to the end user, they are tracked as Tasks rather than User Stories.

Jira Issue Architecture

- **Epic:** Cloud Database Migration
- **User Story:** As a customer, I want faster page load times during checkout (Business value).
- **Task (Example below):** Performance benchmark testing for AWS Aurora instance (Technical requirement).
- **Sub-task:** Write automated load-testing scripts.
- **Sub-task:** Execute 10,000 concurrent user simulation.

Real-World Example of the Created Jira Task

Jira Field	Value / Configuration
Project	E-Commerce Platform (EC)
Issue Type	Task
Summary	Execute performance benchmarking on AWS Aurora staging database
Description	**Objective:** Validate that the new AWS Aurora cluster can handle peak Black Friday traffic profiles before we migrate production data. **Scope of Work:** 1. Spin up an isolated Aurora replica in the staging VPC. 2. Run the existing Locust load-testing suite against the database instance. 3. Capture CPU utilization, read/write IOPS, and query latency spikes. **Definition of Done (DoD):** • Load test run continuously for 4 hours at 2x normal peak load. • Performance report generated and attached to this ticket. • Query latency remains under 200ms at peak.
Priority	High
Assignee	Sarah Jenkins (Lead DevOps Engineer)
Labels	devops, database-migration, testing
Sprint	Q4_Migration_Sprint_2
Linked Issues	is blocked by EC-405: Deploy Aurora CloudFormation Templates relates to EC-102: Cloud Database Migration (Epic)

How to write Automation Rules to automatically spawn sub-tasks when this task is created

The Goal

Whenever a standard task with the summary "Execute performance benchmarking on AWS Aurora staging database" is created, Jira will automatically generate two sub-tasks:

- Write automated load-testing scripts.
- Execute 10,000 concurrent user simulation.

Step-by-Step Automation Setup

1. Navigate to Automation

Open your Jira Project, click on Project settings in the sidebar, and select Automation.

2. Create a New Rule

Click the Create rule button in the top right corner.

Trigger: When the main Task is created

- Select Issue created as your trigger.
- Click Save.

Condition: Ensure it only runs on our specific task

Click New component and choose Jira service management condition or standard Issue fields condition. Configure the condition to isolate your target task:

- Field: Issue Type | Condition: equals | Value: Task
- (Optional) Add a second condition for the text: Field: Summary | Condition: contains | Value: performance benchmarking